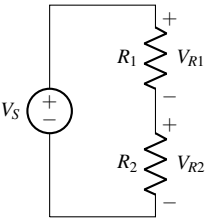
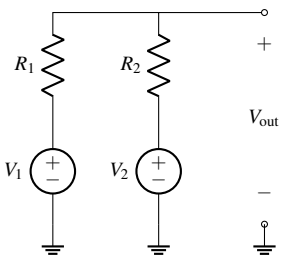
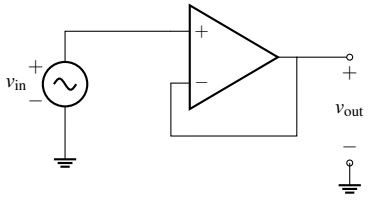
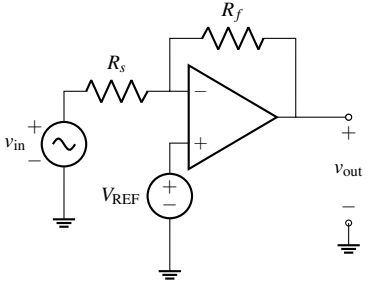
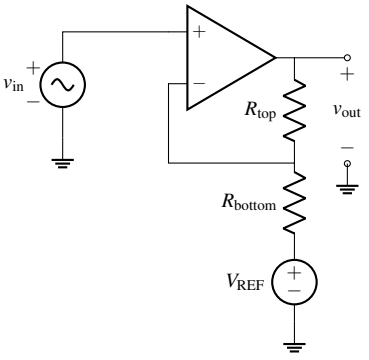


EECS 16A Designing Information Devices and Systems I

Discussion 11A

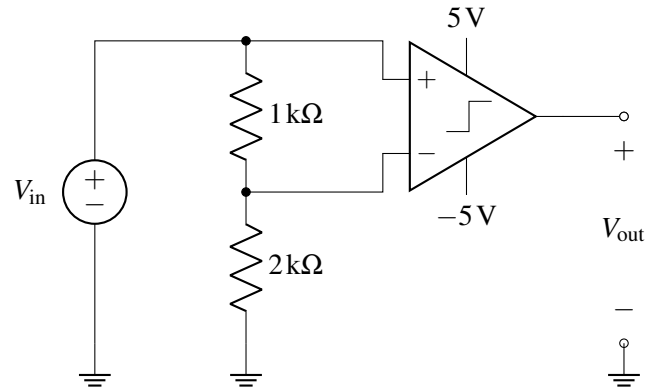
For Reference: Example Circuits

Voltage Divider  $V_{R2} = V_S \left(\frac{R_2}{R_1 + R_2} \right)$	Voltage Summer  $V_{out} = V_1 \left(\frac{R_2}{R_1 + R_2} \right) + V_2 \left(\frac{R_1}{R_1 + R_2} \right)$	Unity Gain Buffer  $\frac{v_{out}}{v_{in}} = 1$
Inverting Amplifier  $v_{out} = v_{in} \left(-\frac{R_f}{R_s} \right) + V_{REF} \left(\frac{R_f}{R_s} + 1 \right)$	Non-inverting Amplifier  $v_{out} = v_{in} \left(1 + \frac{R_{top}}{R_{bottom}} \right) - V_{REF} \left(\frac{R_{top}}{R_{bottom}} \right)$	

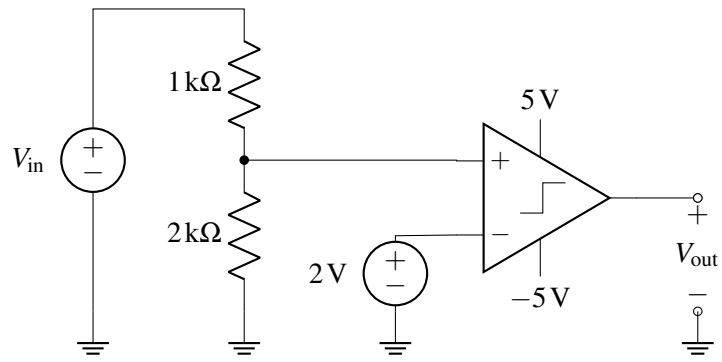
1. Comparators

For each of the circuits shown below, plot V_{out} for V_{in} ranging from -10 V to 10 V for part (a) and from 0 V to 10 V for part (b). Assume $V_{SS} = -V_{DD}$.

(a)

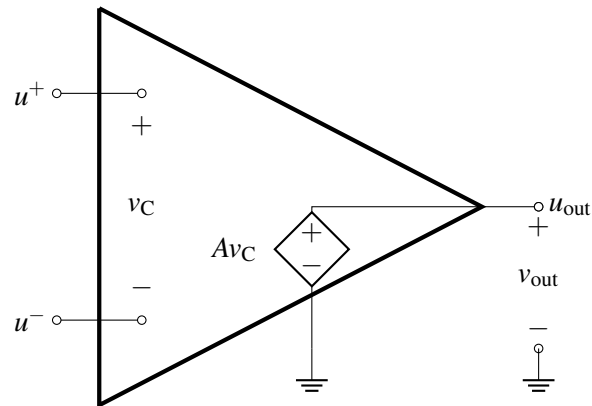


(b)



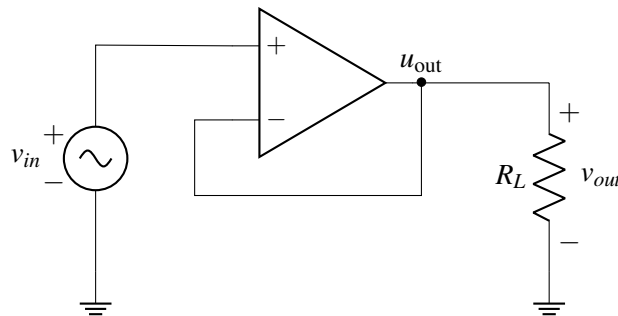
2. Op-Amp Rules and Negative Feedback Rules

Here is an equivalent circuit of an op-amp (where we are assuming that $V_{SS} = -V_{DD}$) for reference:



- What are the currents flowing into the positive and negative terminals of the op-amp (i.e., what are I^+ and I^-)? Based on this answer, what are some of the advantages of using an op-amp in your circuit designs?
- Suppose we add a resistor of value R_L between u_{out} and ground. What is the value of v_{out} ? Does your answer depend on R_L ? In other words, how does R_L affect Av_C ? What are the implications of this with respect to using op-amps in circuit design?

For the rest of the problem, consider the following op-amp circuit in negative feedback:



- Assuming that this is an ideal op-amp, what is v_{out} ?
- Draw the equivalent circuit for this op-amp and calculate v_{out} in terms of A , v_{in} , and R_L for the circuit in negative feedback. Does v_{out} depend on R_L ? What is v_{out} in the limit as $A \rightarrow \infty$?